

Brayton Hall

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Data Scientist | Data Engineer

Experience in data acquisition and modeling, statistical analysis, machine learning, and natural language processing. With a background in mathematical logic and philosophy at Virginia Tech, I excel at **communicating insights on technical, high dimensional, and structured or unstructured data in a way that is interpretable and precise**. My experience in data science across the medical and banking industries has given me versatility through exposure to a massive variety of data, in regard to both data dimensionality and business goals.

TECHNICAL SKILLS

Machine Learning, Applied Statistics, SQL, Python, Docker, PySpark, Hadoop, scikit-learn, NumPy, Pandas, Keras, Regression, Classification, Clustering, Seaborn, matplotlib, Git, NLTK, Gensim, Word2Vec, BeautifulSoup, spaCy, Heroku, Streamlit, React, HTML, CSS, Javascript, DBMS including Spark, Impala, Hive, and **state-of-the-art transformer models** such as BERT, GPT and T5

EMPLOYMENT HISTORY

Data Scientist, **Bank of America**, Charlotte, North Carolina 01/2022 - Current

- Develop NLP models for text classification and insight on high volumes of text data from automatic speech recognition (ASR) engines, on the main DS team driving a \$100 million cost reduction project for automating complaint capture
- Represent the core data science team to deliver interpretable reports to executive leadership, using SQL and state-of-the-art transformer models such as BERT, GPT, and T5 in combination with rule-based lexicon strategies in multiple database management systems (e.g., Spark, Impala, and Hive)

Data Analyst, **PrimeNeuro**, Durham, North Carolina 10/2020 - 04/2021

- Used machine learning algorithms on fMRI brain-imaging data to predict emergent behavior, with the aim of early interventions for autism by creating behavioral classification models from high-dimensional data
- Cleaned, visualized, and modeled image and time-ordered data using AWS S3 and EC2, SQL, Docker, Python, Jupyter, and statistical methods such as principal component analysis to extract features from large data sets

Graduate Philosophy and Statistics Instructor, **Virginia Tech**, Blacksburg, Virginia 08/2015 - 08/2019

- Lectured a 150-person seminar, created and maintained a grading database for professors, taught and attended seminars focused on logic, statistical inference, the ethics of science and technology, and the philosophy of natural science
- Taught statistics and math post-graduation, with a focus on applied concepts such as confidence intervals, t-tests, hypothesis testing, and A/B testing, linear algebra, probability theory, Bayesian statistics, and linear regression

TECHNICAL PROJECTS

Semantic Search Engine - [Github](#)

Created a 'search between the lines' app to search by connotations and misremembered quotes rather than exact fragments, inspired by the frustration of simple search engines on e-readers and websites

- Used Doc2Vec (a neural net package based on Word2Vec) in Gensim to vectorize 60,000 paragraphs from free ebooks on Project Gutenberg, and used them to train a model which can infer meaning based on cosine similarity
- Cleaned, organized, and visualized the data (1.2 million words) using Pandas and Seaborn, and stored the model and DataFrame on an S3 bucket with AWS, deployed using Docker on Heroku

COVID-19 Twitter Topic Modeling - [Github](#)

Using a combination of natural language processing packages to find intelligible topic clusters on Twitter

- Scraped 100,000 'coronavirus' tweets using Twint and preprocessed them with Gensim, NLTK, and spaCy
- Used the Python package vaderSentiment to analyze tweets for non-neutral sentiments
- Created a Latent Dirichlet Allocation (LDA) model with Gensim to discover that five distinct Twitter communities emerged: 'Trump', 'Economy', 'Employment', 'Public Health', and 'UK/Boris Johnson'

EDUCATION

Flatiron School, New York, NY 01/2020 - 05/2020

Immersive Data Science Program

- *Relevant experience:* Python, SQL, Statistics, Machine Learning, Computer Science, Data Visualization

Virginia Tech, Blacksburg, Virginia 08/2015 - 05/2017

MA in Philosophy

- *Relevant coursework:* Metalogic and Number Theory, Philosophy of Science, Ethics, Philosophy of Language

UNC Chapel Hill, Chapel Hill, North Carolina 08/2009 - 05/2013

BS in Psychology

- *Relevant coursework:* Statistics, Decision Theory, Calc I, Calc II, Symbolic Logic, Physics, Cognitive Psychology